

An Open-source and Optimization-based AC/DC Power Flow Model for the Brazilian Interconnected Power System

Marck Llerena Velasquez[†], Marcos J. Rider[†] and Daniel Dotta[†]

[†]*State University of Campinas (UNICAMP), Campinas, Brazil*
marck.llerena-velasquez@nrl.gov, dottad@unicamp.br

Abstract—This paper introduces an open-source and optimization-based integrated AC/DC power flow model designed to analyze complex hybrid power grids, such as the Brazilian Interconnected Power System (BIPS). The BIPS is one of the biggest and most complex power grids that interconnects five regions and has high presence of Flexible AC Transmission System (FACTS) devices, including the Thyristor Controlled Series Capacitor (TCSC) and Static Var Compensator (SVC), and bipolar point-to-point line-commutated converters (LCC). The model is formulated as a nonlinear programming (NLP) problem, combining power balance equations, detailed steady-state LCC and FACTS devices equations. In this implementation, the only explicit inequality constraints are those associated with converter station transformer tap limits and delay/extinction angles. All remaining device and network equations are enforced through equality constraints. To handle possible SVC saturation, a post-solution iteration is run to verify reactive limit violations and update each saturated device to its corresponding limit region. The proposed model was implemented in Python using the Pyomo optimization framework and solved using the Ipopt solver and validated with the Anarede software, on a 247-bus BIPS equivalent and further scaled through a large-scale 7308-bus BIPS model. Results show consistent operating points with expected numerical differences due to different assumptions and solution strategies, while demonstrating robust convergence for large-scale AC/DC systems with complex technology integration.

Index Terms—FACTS, hybrid AC/DC systems, large-scale power systems, LCC-HVDC, nonlinear programming, open-source software, static var compensator, thyristor-controlled series capacitor.

I. INTRODUCTION

Power flow analysis is a fundamental process in power systems operations and planning, since it provides the steady-state voltages, angles, and power injections used in security assessment and grid studies [1]. For many years, this process has been solved mainly with Newton Raphson and Fast Decoupled methods, which offer good computational performance for conventional AC networks. As modern grids evolve, however, the modeling scope has expanded beyond the traditional AC setting. The growing integration of FACTS devices and HVDC systems introduces high nonlinearities, which increases the complexity of the power flow problem [2]. In hybrid AC/DC systems, representations are separated into sequential and integrated formulations. Sequential methods solve AC and DC subsystems separately and exchange boundary values

until convergence [3]. This structure reduces matrix size per iteration and is attractive for large systems, but convergence can be weak in stressed or poorly conditioned cases, especially when many controls and limits are active [4].

Integrated methods solve AC and DC equations together in a single model, which improves consistency between network states and converter controls [5]. In this same approach, FACTS devices can also be embedded in the power flow equations through steady-state injection-based representations, enabling their control actions and limits to be handled within the unified network solution [6].

To increase robustness, recent studies have moved toward NLP solvers to address this issues. Compared with traditional approaches, NLP formulations are more flexible for adding limits and control equations. A common approach is the use of soft constraints with slack variables and penalty terms, which helps maintain convergence when strict feasibility is hard to satisfy in stressed operating conditions [7].

Even with these advances, there are still few studies applying this level of detailed modeling to very large systems such as the BIPS. This motivates the present work, an integrated optimization-based AC/DC power flow model with detailed LCC-HVDC and FACTS representation, solved simultaneously and validated at both equivalent and full BIPS scale. The goal is to combine modeling realism with robust convergence for practical large-scale studies. The main contributions are summarized as follows:

- (i) Development and open-source implementation of an integrated nonlinear AC/DC power flow model that explicitly represents FACTS and LCC-HVDC steady-state systems behavior, solved with the Ipopt solver [8] in a single NLP framework with set controls and constraints to model real system operations.
- (ii) Validation of the proposed model with the Anarede software, a Brazilian power system analysis tool [9], using both a system equivalent and the large-scale BIPS model, showing numerically consistent operating points and robust convergence for large-scale cases.

II. PROBLEM FORMULATION

The AC/DC power flow model is formulated as a NLP optimization problem. While traditional root-finding methods solve a rigid system of non-linear equations, this optimization-based approach models physical system limits, such as voltage

security margins and generator reactive power capabilities, directly as inequality constraints within the core network solution [10]. The algebraic structure of the problem consists of continuous decision variables bounded by technical limits, governed by a set of equality and inequality constraints that represent the unified AC and DC network physics.

A. Objective Function

In an optimization-based power flow, the objective function $f(\mathbf{x})$ can be defined as a constant to find any feasible point. In this work, to provide a clear convergence direction for the solver, the objective function is defined as the squared sum of all generator active powers [11]:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{g \in \mathcal{G}} (p_g)^2 \quad (1)$$

subject to:

$$\mathbf{g}(\mathbf{x}) = 0, \quad (2)$$

$$\mathbf{h}(\mathbf{x}) \leq 0. \quad (3)$$

Where $\mathbf{x}$ is the vector of state variables, $\mathcal{G}$ is the set of all generators in the system, and p_g represents the active power injection of generator g . The equality constraints $\mathbf{g}(\mathbf{x})$ represent the complete set of AC and DC power balance equations. The inequality constraints $\mathbf{h}(\mathbf{x})$ enforce operational limits, such as tap limits and firing/extinction angles for converter stations.

B. Optimization-based AC Power Flow

In the present work, a power balance formulation in polar coordinates is adopted. The resulting steady-state equality constraints for active and reactive power balance use compact aggregated terms to represent network injections. For every bus $i \in \mathcal{N}$, where $\mathcal{N}$ represents the set of all system buses, these terms are defined as:

$$P_i^\ell := \sum_{(k,i,j) \in \mathcal{L}_i^{out}} p_{kij}^{fr} + \sum_{(k,j,i) \in \mathcal{L}_i^{in}} p_{kji}^{to}, \quad (4)$$

$$Q_i^\ell := \sum_{(k,i,j) \in \mathcal{L}_i^{out}} q_{kij}^{fr} + \sum_{(k,j,i) \in \mathcal{L}_i^{in}} q_{kji}^{to}, \quad (5)$$

$$P_i^{csc} := \sum_{(m,i,j) \in \mathcal{C}_i^{out}} p_{mij}^{csc} - \sum_{(m,j,i) \in \mathcal{C}_i^{in}} p_{mji}^{csc}, \quad (6)$$

$$Q_i^{csc} := \sum_{(m,i,j) \in \mathcal{C}_i^{out}} q_{mij}^{csc} + \sum_{(m,j,i) \in \mathcal{C}_i^{in}} q_{mji}^{csc}, \quad (7)$$

$$P_i^{dc} := \sum_{h \in \mathcal{H}_i^r} P_r^{ac,h} - \sum_{h \in \mathcal{H}_i^i} P_i^{ac,h}, \quad (8)$$

$$Q_i^{dc} := \sum_{h \in \mathcal{H}_i^r} Q_r^{ac,h} + \sum_{h \in \mathcal{H}_i^i} Q_i^{ac,h}, \quad (9)$$

$$Q_i^{svc} := \sum_{s \in \mathcal{S}_i} q_s^{svc}. \quad (10)$$

Using these aggregated boundary flows, the active and reactive power mismatches are enforced at each bus $i \in \mathcal{N}$ by:

$$\sum_{g \in \mathcal{G}_i} p_g - \sum_{d \in \mathcal{D}_i} p_d - P_i^\ell - P_i^{csc} - P_i^{dc} - g_i^{sh} v_i^2 = 0, \quad (11)$$

$$\sum_{g \in \mathcal{G}_i} q_g - \sum_{d \in \mathcal{D}_i} q_d + b_i^{sh} v_i^2 + Q_i^{svc} - Q_i^\ell - Q_i^{csc} - Q_i^{dc} = 0. \quad (12)$$

The branch active and reactive flows, p_{ij} and q_{ij} , between terminal buses i and j on line $k \in \mathcal{L}$ are calculated as:

$$p_{ij} = g_{ij} \frac{v_i^2}{t_{ij}^2} - \frac{v_i v_j}{t_{ij}} (g_{ij} \cos(\theta_{ij} + \phi_{ij}) + b_{ij} \sin(\theta_{ij} + \phi_{ij})), \quad (13)$$

$$q_{ij} = -B_{ij} \frac{v_i^2}{t_{ij}^2} + \frac{v_i v_j}{t_{ij}} (g_{ij} \sin(\theta_{ij} + \phi_{ij}) - b_{ij} \cos(\theta_{ij} + \phi_{ij})). \quad (14)$$

The transformer tap ratios (t_{ij}) and phase shifts (ϕ_{ij}) in (13) and (14) are modeled as fixed parameters read from the input data. These values represent fixed-tap transformers or on-load tap changers whose operating points are defined before solving the power flow. Bus shunt elements are explicitly represented in the nodal balances in (11) and (12) through g_i^{sh} and b_i^{sh} , while line shunt effects are included in the branch-flow terms. To preserve the behavior of a conventional power flow, selected state variables are fixed according to bus type by setting equal lower and upper bounds at their specified values.

C. Thyristor Controlled Series Capacitor (TCSC)

The impact of these devices on steady-state operation has been extensively documented, highlighting their capacity to control the power transfer on the line by acting on the terminal voltage magnitudes, the phase angles, or the line reactance [12]. In this work, the TCSC is modeled operating exclusively in constant reactance control mode. For each device $m \in \mathcal{C}$ connected between terminal buses i and j , the active and reactive power expressions can be written as:

$$p_m^{csc} = \frac{v_i v_j}{x_m^e} \sin(\theta_i - \theta_j), \quad \forall m \in \mathcal{C}, \quad (15)$$

$$q_m^{csc} = \frac{v_i^2}{x_m^e} - \frac{v_i v_j}{x_m^e} \cos(\theta_i - \theta_j), \quad \forall m \in \mathcal{C}, \quad (16)$$

where v_i and v_j are the terminal voltage magnitudes, $(\theta_i - \theta_j)$ is the phase angle difference, and x_m^e is the device specified reactance. These power transfer terms, p_m^{csc} and q_m^{csc} , are added to the nodal power balance equations (11) and (12), respectively.

D. Static VAR Compensator (SVC)

The reactive power injected or absorbed by the SVC is represented by a piecewise-linear function that captures both the control mode and the specific operating region of the device, these configurations are shown in Figure 1. Depending on whether the SVC $s \in \mathcal{S}$ is operating in reactive power control ($c_s^{svc} = 0$) or injected current control ($c_s^{svc} = 1$), different expressions are applied to characterize the interaction with the AC network, as defined in the Anarede functional

specifications [12]. The reactive power injection q_s^{svc} is formulated as follows:

$$q_s^{svc} = \begin{cases} (v_s^{ref} - v_s^{svc})v_s^t/m_s^{ctrl}, & \text{if } c_s^{svc} = 0, r_s^{gn} = 0 \\ (v_s^{ref} - v_s^{svc})/m_s^{ctrl}, & \text{if } c_s^{svc} = 1, r_s^{gn} = 0 \\ q_s^{\min} \cdot (v_s^t)^2, & \text{if } r_s^{gn} = +1 \\ q_s^{\max} \cdot (v_s^t)^2, & \text{if } r_s^{gn} = -1 \end{cases} \quad \forall s \in \mathcal{S}, \quad (17)$$

where m_s^{ctrl} represents the control slope (droop) of the compensator, and r_s^{gn} is a region indicator variable identifying if the device has reached its inductive (+1) or capacitive (-1) susceptance limits. In this formulation, v_s^{ref} is the target control reference voltage for the specified bus, v_s^{svc} is the voltage at the controlled bus where the SVC is connected, v_s^t represents the terminal voltage of the SVC device, and $q_s^{\min}, q_s^{\max}$ are the minimum inductive and maximum capacitive reactive power operation limits of the compensator, respectively.

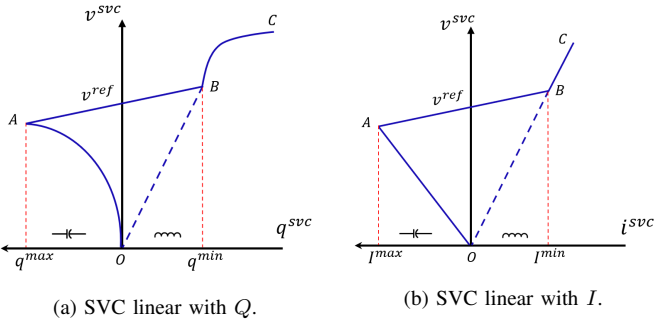


Fig. 1. SVC control modes configurations.

E. Line-Commutated Converter (LCC)

The steady-state modeling of LCC-HVDC systems used in this work follows classical converter relationships, including converter-side voltages, current, control angles, and commutation effects [13], [14]. These relationships provide the analytical basis for representing rectifier and inverter operation in AC/DC power flow studies as shown in Fig. 2. The LCC system is modeled using a unified set of equations for both rectifier ($k = r$) and inverter ($k = i$) terminals [15]. For each pole $h \in \mathcal{H}$, where $\mathcal{H}$ represents the set of all active DC poles, the DC voltage $V_k^{dc,h}$ is expressed as the ideal no-load voltage term plus the commutation reactance drop:

$$V_k^{dc,h} = N_k^h \left(\frac{3\sqrt{2}}{\pi} \frac{a_k^h V_k^{ac,h}}{T_k^h} \cos \theta_k^h + s_k \frac{3X_k^h}{\pi} I_k^{dc,h} \right), \quad \forall h \in \mathcal{H}, k \in \{r, i\}, \quad (18)$$

where $s_r = -1$ for the rectifier and $s_i = +1$ for the inverter, which physically accounts for the polarity change and internal voltage drop across the bridges under commutation. The parameter θ_k^h denotes the operational control angle mapped to each terminal mode; specifically, $\theta_r^h = \alpha_r^h$ represents the firing delay angle for the rectifier, whereas $\theta_i^h = \gamma_i^h$ represents the minimum extinction angle for the inverter. Furthermore, N_k^h

is the number of bridges that the converter station possesses for pole h at terminal k , a_k^h is the transformer ratio, T_k^h is the tap ratio, X_k^h is the commutation reactance, and $I_k^{dc,h}$ is the DC current flowing through the link.

The commutation process induces a phase displacement between the AC voltage and fundamental AC current. The corresponding overlap angle μ_k^h and power factor angle ϕ_k^h for each pole $h \in \mathcal{H}$ and terminal $k \in \{r, i\}$ are governed by:

$$\mu_k^h = \cos^{-1} \left(\cos \theta_k^h - \frac{\sqrt{2} I_k^{dc,h} X_k^h T_k^h}{a_k^h V_k^{ac,h}} \right) - \theta_k^h, \quad (19)$$

$$\phi_k^h = \tan^{-1} \left(\frac{\sin(2\theta_k^h + 2\mu_k^h) - \sin(2\theta_k^h) + 2\mu_k^h}{\cos(2\theta_k^h) - \cos(2\theta_k^h + 2\mu_k^h)} \right). \quad (20)$$

To guarantee mathematical and physical consistency within the NLP optimization framework, the argument of the inverse cosine in (19) is strictly constrained to the domain $[-1, 1]$, preventing non-convergence during high DC current or depressed AC voltage states.

The RMS AC current $I_k^{ac,h}$ at the converter transformer is proportional to the DC current, while the reactive power variable $Q_k^{ac,h}$ at the AC interface is derived for all $h \in \mathcal{H}$ and $k \in \{r, i\}$ as follows:

$$I_k^{ac,h} = \frac{\sqrt{6}}{\pi} N_k^h I_k^{dc,h}, \quad (21)$$

$$Q_k^{ac,h} = \sqrt{3} I_k^{ac,h} (a_k^h V_k^{ac,h} / T_k^h) \sin \phi_k^h. \quad (22)$$

In the nodal balance, the active power terms $P_r^{ac,h}$ and $P_i^{ac,h}$ are treated as fixed setpoints specified by the system data models, representing a withdrawal at the rectifier bus and an injection at the inverter bus, respectively. Conversely, because line-commutated converters inherently consume reactive power regardless of power flow direction, both $Q_r^{ac,h}$ and $Q_i^{ac,h}$ are modeled strictly as reactive power withdrawals. These variables are directly integrated into the nodal power balance equations (11) and (12). Finally, operational feasibility and equipment limits are enforced across all components by:

$$T_k^{\min,h} \leq T_k^h \leq T_k^{\max,h}, \quad \forall h \in \mathcal{H}, k \in \{r, i\}, \quad (23)$$

$$\alpha_r^{\min,h} \leq \alpha_r^h \leq \alpha_r^{\max,h}, \quad \forall h \in \mathcal{H}, \quad (24)$$

$$\gamma_i^{\min,h} \leq \gamma_i^h \leq \gamma_i^{\max,h}, \quad \forall h \in \mathcal{H}. \quad (25)$$

III. SIMULATION PROCESS

The proposed model is implemented in Python using the Pyomo optimization framework and solved via the Ipopt solver. The simulation starts by parsing Anarede files (.pwf) into a readable open-source format (.dat), followed by the execution of the AC/DC power flow model. The post-optimization phase involves verifying SVC operational status; if reactive power limits are violated, the algorithm iteratively adjusts setpoints according to the control logic defined in (II-D) and solves the system again. This ensures that the final operating point is physically feasible and is within the device saturation limits. Validation is then performed by comparing the proposed model results against Anarede. Numerical differences are expected, since the software uses Newton Raphson based algorithms. These differences are particularly evident near operational

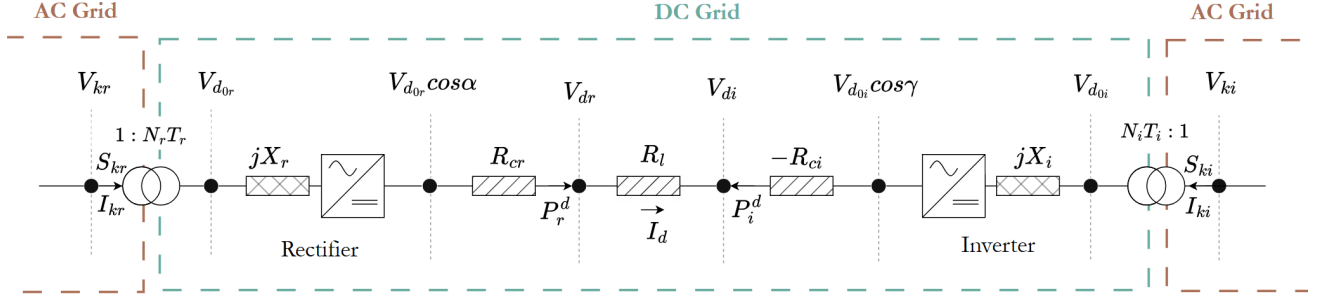


Fig. 2. LCC-HVDC converter stations and circuit representation.

boundaries, such as converter transformer tap limits or maximum firing angles ($\alpha_{max}, \gamma_{max}$), where the two solvers may converge on different but equally feasible points.

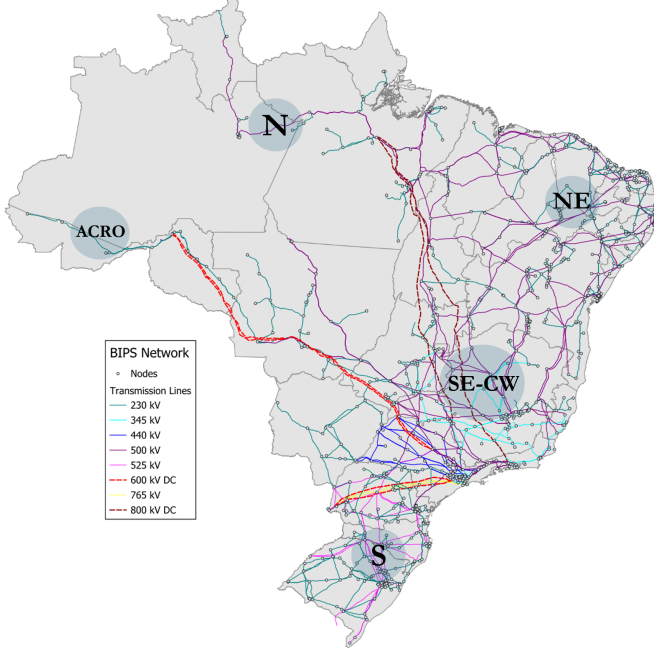


Fig. 3. BIPS transmission network.

IV. NUMERICAL RESULTS

A. BIPS Data Model

The data model used to represent the BIPS transmission network, as seen in Figure 3, consists of 7308 buses, 6477 lines, 3861 two-winding transformers, 68 CSCs, 44 SVCs, 321 line shunts, 1961 bus/line capacitor/reactor banks, 1299 generators and 2736 loads. The operating snapshots are based on cases released by the Brazilian Independent System Operator [16].

As part of this study, the main LCC-HVDC corridors currently relevant to BIPS operation are explicitly modeled. These include the Itaipu link (Foz do Iguaçu–Ibiúna), the Madeira complex (Santo Antônio–Araraquara, ABB and Areva bipoles), and the Belo Monte transmission link (Xingu–Estreito and Xingu–Terminal Rio). In total, the implemented

TABLE I
LCC LINE PARAMETERS IN THE BIPS.

Line Name	Voltage (kV)	Power (MW)	Pole No.	α/γ (°)
Itaipu	600/587.7	1050/1018	4	15/17
Madeira 1	600/557.2	1500/1393	2	15/17
Madeira 2	600/560.4	1500/1401	2	13/18
Belo Monte 1	800/765.2	2000/1913	2	14.3/15
Belo Monte 2	800/759.5	2000/1899	2	18.4/17

set comprises 12 poles organized as three major LCC schemes: four poles at 600 kV for Foz/Ibiúna, four poles at 600 kV for Santo Antônio/Araraquara, and four poles at 800 kV for the Xingu corridors, with nominal pole powers in the range of approximately 1566–2000 MW. Their representative operating settings are summarized in Table I.

B. Model Validation

To validate the proposed AC/DC power flow model, the corresponding analysis is performed on the equivalent BIPS model focusing on the point-to-point DC configuration on the LCC-HVDC links described on Table I. Using a flat start configuration on the AC side, and keeping DC voltage constant on one of each converter stations, the results for the equivalent BIPS are shown in Fig. 4 and Fig. 5, respectively.

It can be observed that the proposed AC/DC power flow model tracks the AC voltage magnitudes, phase angles, generator active and reactive power outputs, branch flows, and the control setpoints of the FACTS and LCC-HVDC devices across the equivalent and full BIPS network. To quantify the similarity between the proposed AC/DC model and Anarede, an R^2 coefficient of determination was applied across the validation datasets, confirming a high fit on the results.

The mismatches observed when comparing the results with Anarede are primarily driven by differences in core modeling approaches and power flow assumptions. Because a NLP problem of this scale possesses multiple local solutions, the solver successfully converges to a physically feasible operating point around the feasible set.

Furthermore, minor variations in the converter firing (α) and extinction (γ) angles are expected. While the acceptable operational limits for these angles typically span from 5° to 90°, in steady-state operation it is preferred to have small angle

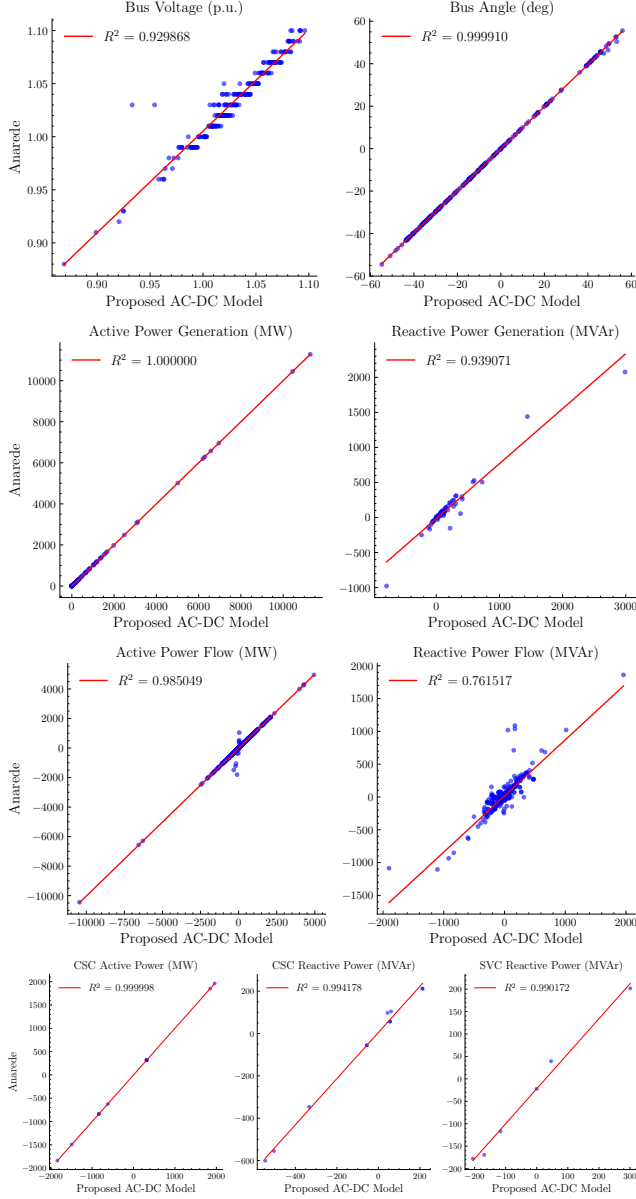


Fig. 4. Validation of equivalent BIPS model at AC components.

values for safe operation and to avoid commutation failure as shown in Table I; in this simulation, the control angles variation was in a range from 15° to 23° . Because these parameters are handled within an optimization framework, the solver's primary objective is to ensure they converge within coherent operational boundaries rather than matching the exact value. Specifically, the NLP formulation evaluates these variables from a system-wide perspective, balancing active and reactive power constraints across the entire network and optimizing the control angles to achieve a feasible solution.

Unlike the active power results, reactive power are highly sensitive to small network modeling variations. In the proposed formulation, the transformer model is simplified by applying a tap ratio configuration directly to the power flow equations. The bus and line shunts are directly added into the bus susceptibility (B) values. These simplified representations can modify

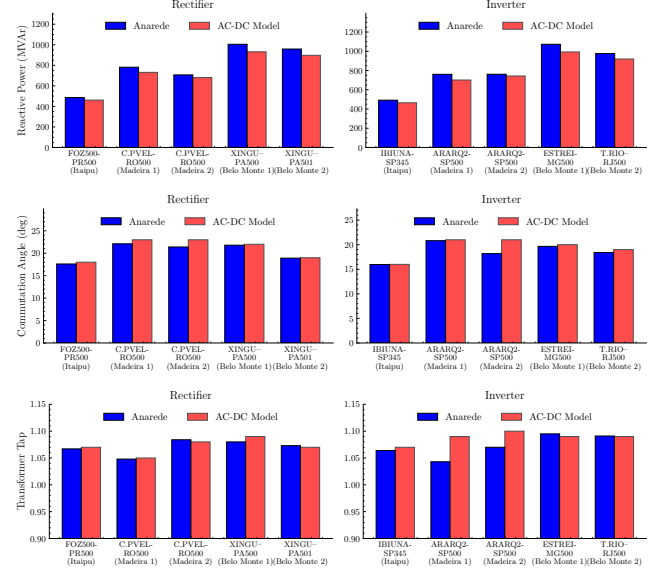


Fig. 5. Validation of equivalent BIPS model at DC components.

the localized reactive power distribution across the network. Consequently, mismatches can be observed when comparing the results with a commercial software like Anarede.

To evaluate the performance and scalability of the proposed AC/DC power flow model, the 7308-bus BIPS model was simulated under a normal steady-state operation mode for the LCC-HVDC. The resulting power flow profiles demonstrate that the active and reactive generator outputs, AC bus voltage magnitudes and phase angles, FACTS and LCC-HVDC terminal parameters are kept within their operational boundaries. These profiles are observed in Fig. 6 and Fig. 7.

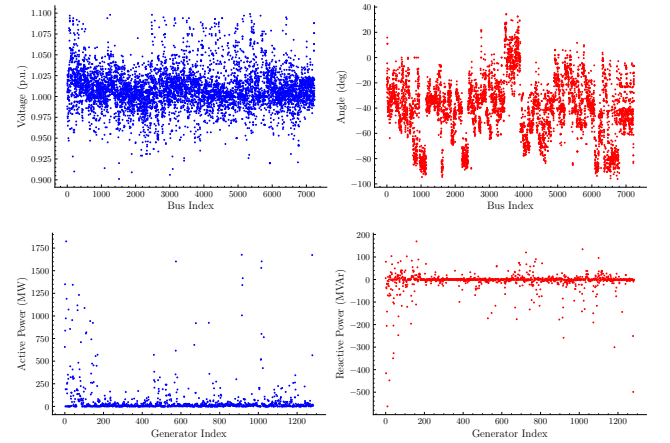


Fig. 6. AC/DC power flow results for buses and generators.

To support the heavy transmission conditions of high active power flows transmitted over the long-distance LCC-HVDC systems, the converter transformer tap positions are controlled within a physical range of 0.9 to 1.2. Keeping the tap ratios in this range is important in normal operating mode, as they adjust the AC side voltage to minimize the converter firing angles (α) and extinction angles (γ), directly minimizing the high reactive power demand consumed by the converter

stations, as can be seen in Fig. 8. Additionally, the highly

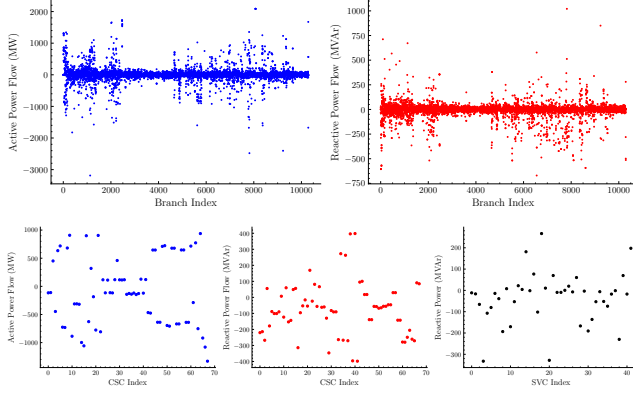


Fig. 7. AC/DC power flow results for branches and FACTS devices.

meshed nature of the BIPS requires active and reactive flow control via FACTS devices, without them the system would not converge to a feasible point. The proposed model incorporates these formulations into the nodal power balance equations. The active and reactive power through the CSC is controlled by active (P_{csc}) and reactive (Q_{csc}) flow constraints, which adjust them based on the equivalent series reactance (X_{esp}). Voltage support and reactive power control are enforced at SVC buses. The model implements piecewise voltage control constraints that govern the reactive power injection (Q_c) based on localized voltage deviations ($V - V_0$) and slope characteristics (I_{nci}), accounting for physical operating limits (Q_{cn}, Q_{cm}). In terms of computational performance, the pro-

posed optimization-based approach requires more processing time as the system scale increases compared to traditional sequential power flow solvers. For the BIPS network, the proposed NLP formulation converges after approximately 80 iterations, requiring around 20 seconds of total execution time. In comparison, the commercial software Anarede solves the

same network configuration in 5 seconds and 10 iterations. This computational tradeoff is expected, as the proposed model solves the full non-linear AC/DC system constraints in a single optimization problem, whereas sequential tools rely on decoupled approaches where the AC and DC networks are solved separately.

V. CONCLUSIONS

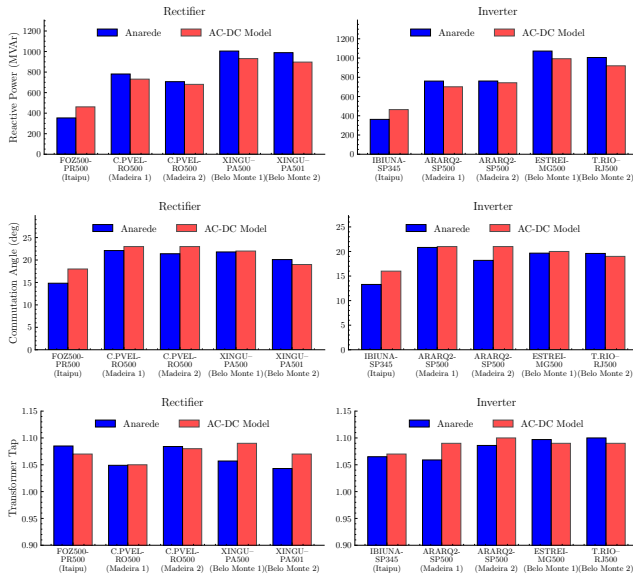
This paper presents an open-source optimization-based, integrated AC/DC power flow model for the steady-state analysis of large-scale power systems incorporating LCC-HVDC and FACTS devices. The proposed model is formulated as a single, NLP problem that simultaneously solves nonlinear AC equations, DC network constraints, and converter control characteristics. The feasibility of the model was demonstrated using both an equivalent and the full BIPS models, and validated against the commercial software Anarede. While numerical differences exist between the proposed model and Anarede due to specific assumptions and component modeling, the results demonstrate that the formulation is a reliable starting point for analyzing the complexity of the BIPS.

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Fig. 8. Validation of BIPS model at DC components.

posed optimization-based approach requires more processing time as the system scale increases compared to traditional sequential power flow solvers. For the BIPS network, the proposed NLP formulation converges after approximately 80 iterations, requiring around 20 seconds of total execution time. In comparison, the commercial software Anarede solves the



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